

Solving Inequalities with the Distributive PropertyAdditional Practice 7A.01.09 **ANSWER KEY**For questions 1 through 4, use the inequality $3(7 - 2h) > 15$.

1. Solve it by using the distributive property first.

$$21 - 6h > 15 \quad (\text{distribute the 3 to both terms})$$

$$-6h > -6 \quad (\text{subtract 21 from both sides})$$

$$h < 1 \quad (\text{divide both sides by } -6 \text{ and REVERSE})$$

2. Solve it again by eliminating the 3 first.

$$7 - 2h > 5 \quad (\text{divide both sides by 3 — no reversal, 3 is positive})$$

$$-2h > -2 \quad (\text{subtract 7 from both sides})$$

$$h < 1 \quad (\text{divide both sides by } -2 \text{ and REVERSE})$$

Both routes reverse exactly once, and both reverse at the same kind of step: dividing by a negative coefficient.

3. Graph the solution on the number line.

*Open circle at 1, shaded left. The symbol is $>$, so 1 itself is not a solution.*

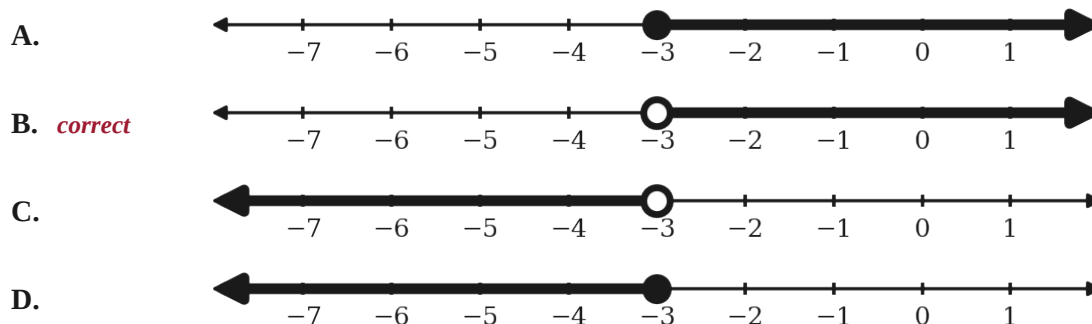
4. Verify your solution set by substitution. Test one value inside it and one value outside it.

Inside: $h = 0 \rightarrow 3(7) = 21$, and $21 > 15$ ✓

Outside: $h = 2 \rightarrow 3(3) = 9$, and $9 > 15$ ✗

Boundary: $h = 1 \rightarrow 3(5) = 15$, and $15 > 15$ ✗ — correct, the boundary is excluded.

5. Which graph represents the solution to
- $-8 < \frac{3}{2}(c - 9)$
- ?



Multiply both sides by $\frac{3}{2}$: $-12 < c - 9$, so $c > -3$.

Choice B. A uses a closed circle for a strict symbol; C and D have the direction reversed. Nothing here reverses — $\frac{3}{2}$ is positive — so a student who flips out of habit lands on C.

For questions 6 through 9, solve using whichever method you prefer, then graph the solution set.

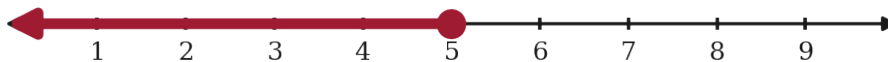
6. $6(g - 2) \leq 18$

$g - 2 \leq 3$ (divide both sides by 6)

$g \leq 5$

Check: $g = 0$ gives $6(-2) = -12$ and $-12 \leq 18$ ✓; $g = 8$ gives $6(6) = 36$ and $36 \leq 18$ ✗

Solution. $g \leq 5$



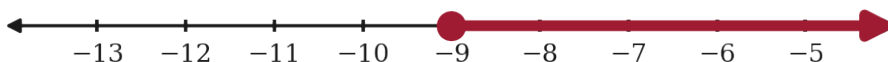
7. $21.6 \leq 7.2(12 + x)$

$3 \leq 12 + x$ (divide both sides by 7.2)

$-9 \leq x$, so $x \geq -9$

Check: $x = 0$ gives $7.2(12) = 86.4$ and $21.6 \leq 86.4$ ✓; $x = -12$ gives 0 and $21.6 \leq 0$ ✗

Solution. $x \geq -9$



$21.6 \div 7.2 = 3$ exactly. Rewrite $-9 \leq x$ as $x \geq -9$ before graphing.

8. $-(5/2)(\frac{3}{4}y + 6) < -22\frac{1}{2}$

$\frac{3}{4}y + 6 > 9$ (multiply both sides by $-2/5$ and REVERSE)

$\frac{3}{4}y > 3$

$y > 4$

Check: $y = 8$ gives $-(5/2)(12) = -30$ and $-30 < -22.5$ ✓; $y = 0$ gives -15 and $-15 < -22.5$ ✗

Solution. $y > 4$



$(-22\frac{1}{2})(-2/5) = 9$. This is the hardest computation on the sheet; the reversal happens once, at the first step.

9. $(3/8)(p + 29) > 9$

$p + 29 > 24$ (multiply both sides by $8/3$ — no reversal)

$p > -5$

Check: $p = 0$ gives $(3/8)(29) \approx 10.9$ and $10.9 > 9$ ✓; $p = -10$ gives $(3/8)(19) \approx 7.1$ and $7.1 > 9$ ✗

Solution. $p > -5$



$9 \div (3/8) = 9 \times (8/3) = 24$.

10. A classmate was absent today. Describe, in words, two different ways to solve $-10 \geq -2.5(z + 8)$. Then give the solution.

Way 1 — divide first. Divide both sides by -2.5 and REVERSE: $4 \leq z + 8$. Subtract 8 from both sides: $-4 \leq z$, so $z \geq -4$.

Way 2 — distribute first. $-2.5(z) + (-2.5)(8) = -2.5z - 20$, so $-10 \geq -2.5z - 20$. Add 20 to both sides: $10 \geq -2.5z$. Divide both sides by -2.5 and REVERSE: $-4 \leq z$, so $z \geq -4$.

Solution: $z \geq -4$

Both ways reverse once, and in both the reversal is caused by the -2.5 . Distributing the -2.5 across the parentheses is not itself a reversal — that step only rewrites one side.